

Experimental Design & Scope of Inference



Observational studies vs. experiments

- 1 A researcher records the exercise habits and cholesterol levels of 500 people without intervening in their lifestyle. Identify whether this is an observational study or an experiment, and explain what type of conclusion (association or causation) it can support.
 - 2 A researcher randomly assigns 200 volunteers to either an exercise program or a no-exercise control group, then measures cholesterol after 6 months. Identify this design, and explain why it can support a causal conclusion.
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The three principles of experimental design (RRCB)

- 3 State the three key principles of a well-designed experiment (sometimes remembered as RRC: Random assignment, Replication, Control of confounding variables), and briefly explain the purpose of each.
 - 4 A study testing a new fertilizer randomly assigns 60 plants to fertilizer or no-fertilizer groups, but only uses 3 plants per group. Identify the design flaw, and explain the consequence.
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Blocking and matched pairs designs

- 5 A study on a new medication expects men and women to respond differently. Describe how a randomized block design would be set up, using sex as the blocking variable.
 - 6 A researcher wants to compare two brands of running shoes on the same 20 runners, having each runner test both brands (in random order) and comparing their times. Identify this design.
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Placebo effect and blinding

- 7 Explain what the placebo effect is, and describe how a single-blind experiment design controls for it.
 - 8 Explain the difference between a single-blind and a double-blind experiment.
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Scope of inference

- 9 A study uses a simple random sample from the population but is purely observational (no random assignment of treatment). State what type of conclusion (about the population, about causation, both, or neither) is justified, and explain why.
- 10 A study randomly assigns treatments to a group of volunteers who were not randomly selected from any larger population. State what type of conclusion is justified.
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Putting it together: evaluating a study design

- 11 A pharmaceutical company recruits 300 volunteers (not randomly selected from the general population) and randomly assigns them to receive either a new drug or a placebo, in a double-blind design. Evaluate what conclusions can and cannot be drawn from this study.
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Confounding variables

- 12 An observational study finds that people who drink more coffee have a higher rate of heart disease. Explain how smoking could act as a confounding variable in this relationship.
- 13 Explain how a well-designed randomized experiment would help rule out a variable like smoking as a confounder in the coffee and heart disease example.
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Identifying experimental units and treatments

- 14 A researcher tests three different fertilizer amounts (low, medium, high) on tomato plants, measuring final yield. Identify the experimental units, the treatment, and the response variable.
- 15 A pharmaceutical researcher is testing whether a new sleep aid reduces the time it takes people to fall asleep, and worries that people with severe insomnia might respond very differently from people with mild insomnia. Design a study using a randomized block design that accounts for this concern, describing each step, and explain the specific benefit this design provides over a simple completely randomized design in this scenario.
- 16 A company tests a new website design by randomly assigning visitors to see either the old design or the new design, without telling visitors which version is the "experimental" one, and measures the percentage who complete a purchase. Explain why this study can support a causal conclusion about the website redesign, and describe the specific scope (which population) that conclusion applies to.

- 17** A university studies whether a new tutoring program improves final exam scores. Volunteers who sign up for tutoring are compared to students who did not sign up, without any random assignment. At the end of the semester, tutored students score noticeably higher on average. Explain why this study cannot establish that the tutoring program caused the improvement, identifying a specific plausible confounding variable.